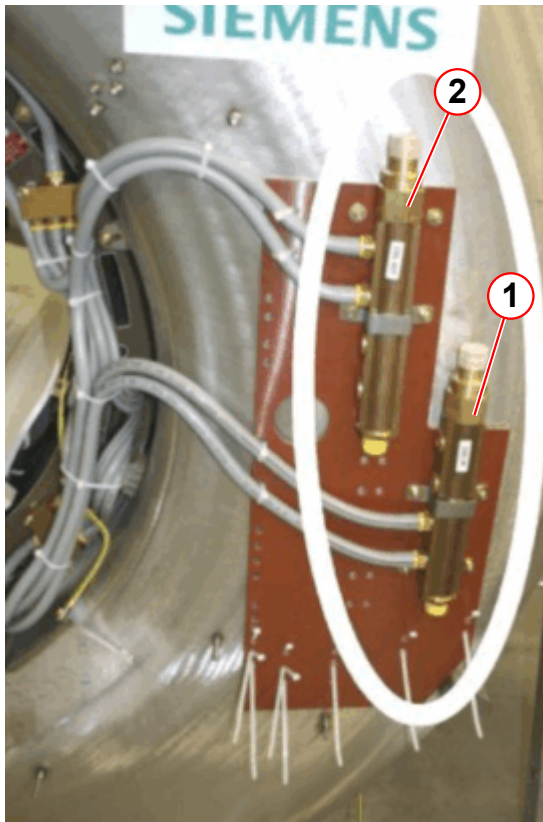


4.1.1.8 Draining the Gradient Coil and Water Hoses

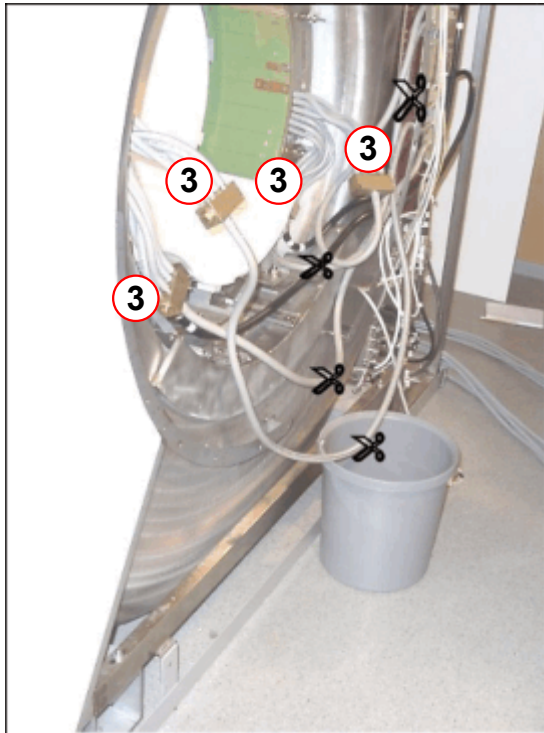
Fig. 110: Main manifolds (hoses disconnected in picture)



(1) IN
(2) OUT

- Quickly disconnect the two main hoses for supply and return (→ 1 and 2/ Fig. 110 Page 120) each from the main manifolds.
- Cut all cable ties that fix the water hoses to the magnet and gradient coil

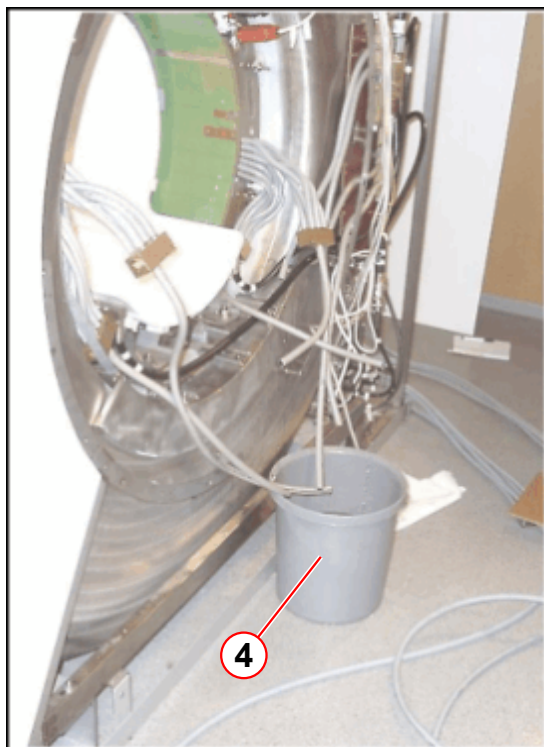
Fig. 111: Cutting the water hoses



(3) Small manifolds

- Detach the four small manifolds (→ 3/Fig. 111 Page 121) from the gradient coil.
- Use the ceramic knife and cut (scissors in (→ Fig. 111 Page 121)) the four water hoses one at a time (supply and return) halfway between the four gradient coil manifolds and the two main manifolds.
 - » Attention: Water will spill as soon as you cut the hoses. Therefore ensure that the old hoses are positioned over the pail

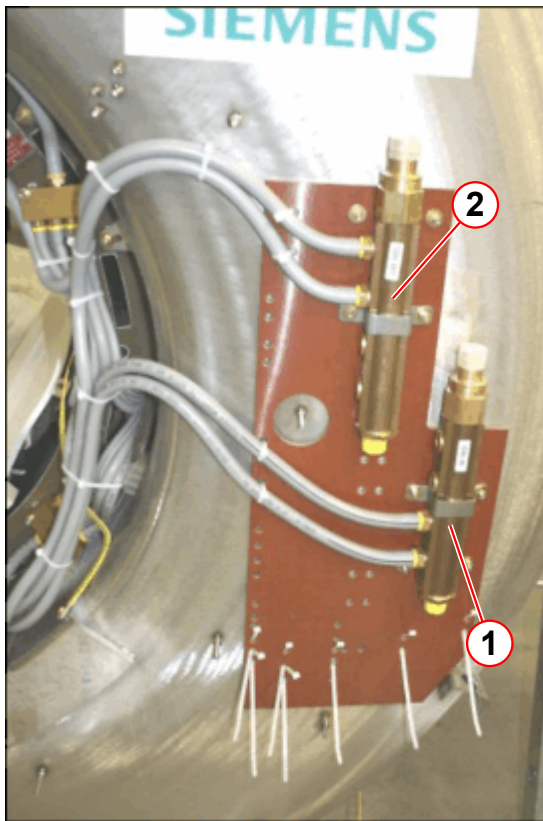
Fig. 112: Draining the gradient coil



(4) Water pail

- Drain the water into the pail (→ 4/Fig. 112 Page 121).

Fig. 113: Main manifolds



- (1) Main supply
(2) Main return

- Detach the two main manifolds for supply and return (→ 1 and 2/Fig. 113 Page 122) from the bakelite board.

4.1.1.9 Removing the four small Manifolds

All 24 gray water hoses (12 to the left and 12 to the right) must be cut between the 4 smaller manifolds and the gradient coil hose nipples

Leave 30 cm of the old hose connected to the gradient coil (→ Fig. 116 Page 123).

Fig. 114: Cutting the water hoses at the grad. coil



- Cut all 24 water hoses (→ Fig. 114 Page 122) with the ceramic knife and remove the old manifold and hose assembly (→ Fig. 115 Page 123).

Fig. 115: Old assembly

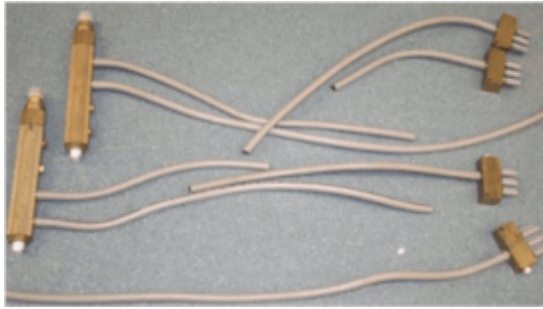


Fig. 116: All 24 water hoses are cut



Once the water hose and manifold (→ Fig. 115 Page 123) have been removed, the remaining 24 hose pieces (→ Fig. 116 Page 123) must be removed from the nipples on the gradient coil (→ Removing the Water Hoses from the Gradient Coil Nozzles / Page 123).

4.1.1.10 Removing the Water Hoses from the Gradient Coil Nozzles

Fig. 117: Perform horizontal cut



The water hoses must be cutoff in a horizontal direction, as demonstrated in (→ Fig. 117 Page 123) and (→ Fig. 118 Page 124).

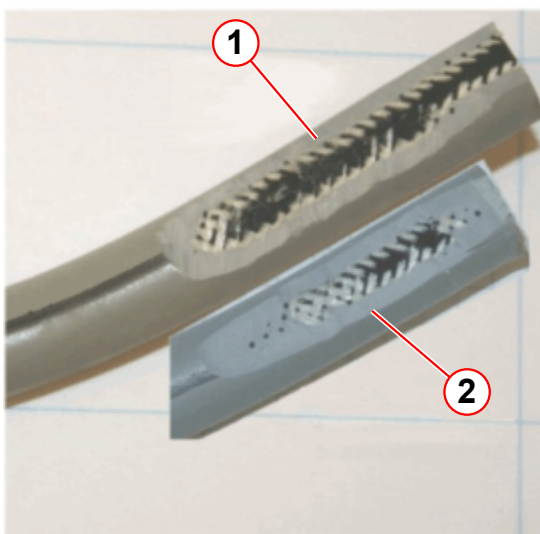
To be able to pull the water hose off the barbed nipples, cut through the Kevlar braiding completely as in (→ Fig. 119 Page 124) and (→ Fig. 120 Page 124). Otherwise, the water hose cannot be pulled off the barbed hose nipple.

Fig. 118: Cutting across the hose



- Carefully guide the ceramic blade across the sleeve on the water hose (→ Fig. 118 Page 124).
- Perform a clean horizontal cut - avoid cutting the brass nipple.
 - » The brass nipple is a delicate component. It must not be scratched or damaged.

Fig. 119: Kevlar braiding



- (1) Good
(2) Not Good

- » It is important to cut through the Kevlar braiding (→ 1 and 2/ Fig. 119 Page 124).

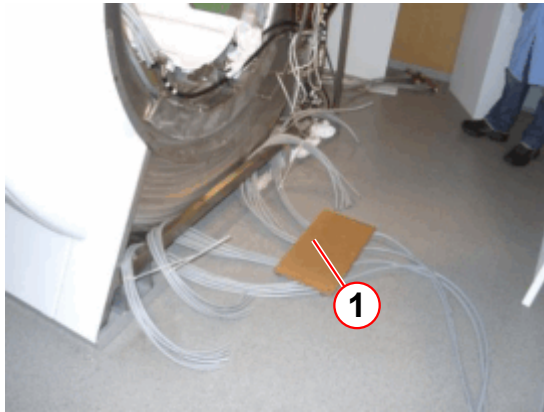
Fig. 120: Kevlar braiding cut through



Once the Kevlar braiding has been cut through, the water hose can be pulled off with ease (→ Fig. 120 Page 124).

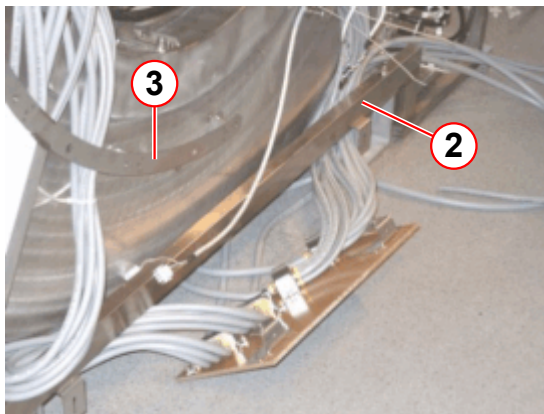
4.1.1.11 Installing the new Manifold

Fig. 121: New manifold assembly



- Spread out the new assembly on the floor(→ 1/Fig. 121 Page 125).

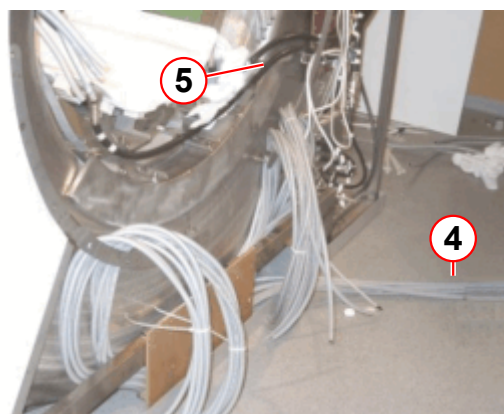
Fig. 122: Mounting board



- Route the new 24 water hoses behind the crossbeam (→ 2/Fig. 122 Page 125) and between the funnel-mounting sword (→ 3/Fig. 122 Page 125) and the magnet to the gradient coil water nipples.
- Simply hang the new mounting board with the four manifolds centered on the crossbar (→ Fig. 122 Page 125) and (→ Fig. 123 Page 126).

(2) Cross beam
(3) Funnel mounting sword

Fig. 123:



(4) Supply and return hoses
(5) RF cables

- Route the 4 supply and return hoses (→ 4/Fig. 123 Page 126) behind the crossbar, the funnel mounting sword and the RF cables (→ 5/Fig. 123 Page 126) up to the two large supply and return manifolds
 - » It is important that the **new** hoses leading to the 2 main manifolds are routed between the magnet and the body coil RF cables for forward and reverse.

The **new** four hoses are labeled with in and out

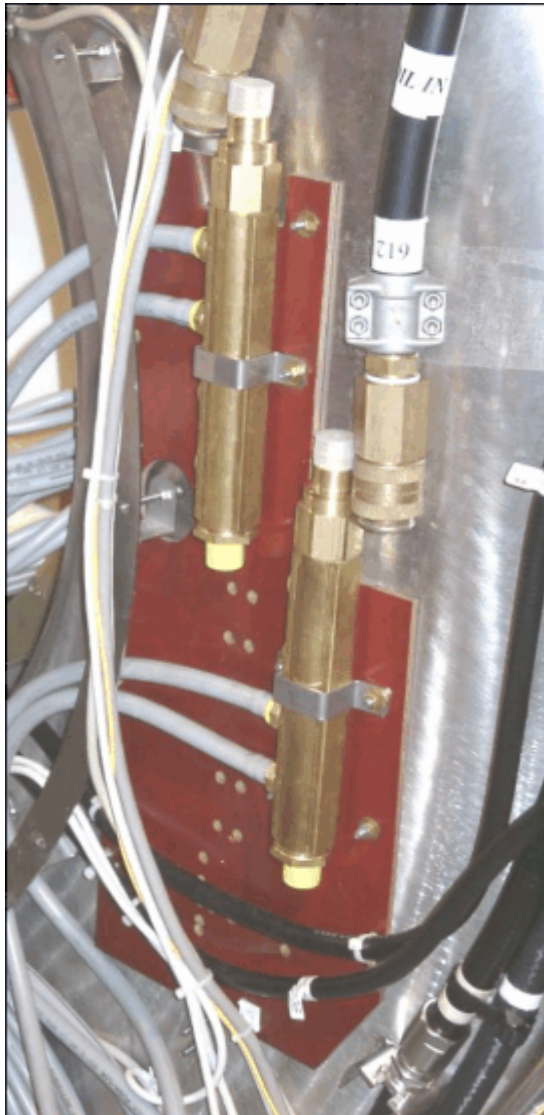
- » At this point, the two new main manifolds should not yet be mounted onto the magnet.

Fig. 124: Lubricated barbed hose nipple



- Lubricate the main manifold hose nipples (→ Fig. 124 Page 126) and connect the water hoses for IN (supply) and OUT (return) by pushing the four hoses over the hose nipples on the manifolds.
 - » Do not mix up the hoses for supply and return

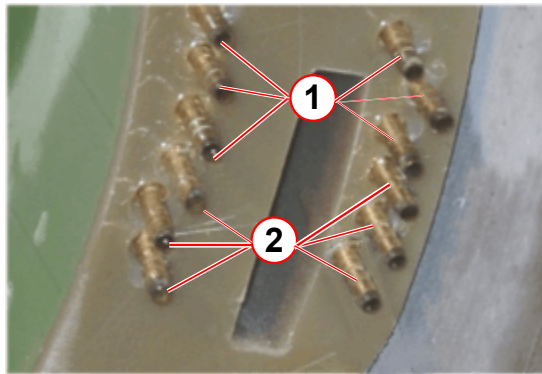
Fig. 125: Main manifolds



- Now fasten the two main manifolds with the U-brackets to the brown bakelite board (→ Fig. 125 Page 127).

4.1.1.12 Fitting the 24 Water Hoses to the Water Nozzles at the Gradient Coil

Fig. 126: Hose nipples on grad. coil



(1) IN
(2) OUT

The method for connecting the hoses to the nozzles is always the same:

Apply lubricant to the barbed hose nipple as in (→ Fig. 124 Page 126) and push the water hose over the barbed brass nipple in one firm motion.

- » The 6 IN water supply hoses go on top and the 6 OUT water return hoses to the bottom.
(→ 1 and 2/Fig. 126 Page 128)

4.1.1.13 Left-hand side at Magnet rear

Fig. 127:



Fig. 128:



- Repeat this procedure for all 12 water hoses for **IN** and **OUT** for the left area.
- The 6 water **IN** hoses should go on top and the 6 water **OUT** hoses go to the bottom.
- Once all 12 hoses have been mated, use cable ties and sticky pads and secure the water hoses to the magnet.
 - » Before attaching the sticky pads, clean the surface with alcohol or similar cleaning agent to remove grease.

The aim is to route the water hoses so that the funnel does not come into contact with water hoses when it is installed later (→ Fig. 128 Page 128) and (→ Fig. 129 Page 129).

Finished with left side

4.1.1.14 Right-hand side at Magnet rear

The procedure is virtually the same as for the left-hand side of the magnet.

Ensure that all water hoses are routed individually and arranged adjacently. Use plenty of sticky pads and cable ties.

- » In the case of water hoses, the top is cold and bottom is warm.

Fig. 129:

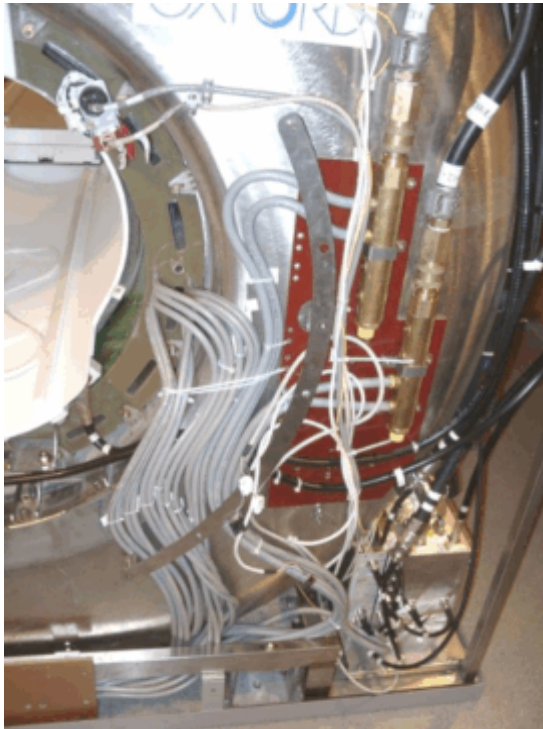
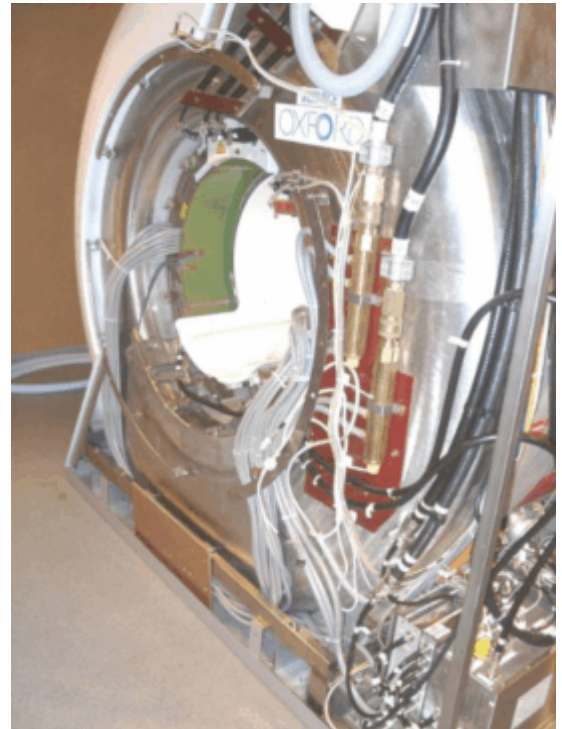


Fig. 130: Modification finished



Arranged water hoses in (→ Fig. 130 Page 129).

- Quick connect the main water hoses for supply and return at the main manifolds.

4.1.1.15 Filling up the Water Reservoir with distilled water

The filling port for the system's cooling water reservoir is on top of the RCA.

System must be switched on

- Remove the plug and fill up the reservoir with distilled water until the yellow light on the Satchwell controller extinguishes.

4.1.1.16 Final Checks

- Go to the back of the magnet and check the entire new water hose section for possible leaks
- All connections should be watertight and dry.

- Finish off your system - put on all covers, close cabinet doors, clean up your trash and tools.

Perform Quality Assurance:

- Coil Check (SN DIP)
- Phantom Shim Check
- Spike Check